openEHR EHRCON26

Modelling the Episode-Oriented Medical Record according to Solon in openEHR

Patterns for Longitudinal, Problem- and Episode-Centred Care

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INTRODUCTION

Electronic health records often organise clinical information around documents, encounters, source systems or isolated problem lists. This limits the visibility of longitudinal clinical trajectories across contacts, providers and care settings.

Solon's episode-oriented medical record instead organises healthcare documentation around episodes of care related to individual health problems.

METHODOLOGY

A design-science approach, supported by a conceptual proof-of-concept implementation, translated Solon's methodology into a conceptual information model and mapped it to openEHR structures including COMPOSITIONs, ENTRYs, FOLDERs, LINKs and archetyped CLUSTERs.

- Three implementation patterns compared using a structured evaluation framework
- Realistic longitudinal patient history with 132 clinical entries
- Eight clinician-centred user stories covering chronological, episode-centred, diagnosis/problem-list and past medical history views

RESULTS

FOLDER-based indexing
Intuitive patient-wide navigation and longitudinal organisation.
Requires governance of lifecycle, naming and update policies.

LINK-based referencing
Flexible explicit cross-composition and cross-entry relationships.
Depends on consistent semantics and increases query/application complexity.

CLUSTER-based relationships
Strongest semantic governance: explicit, coded and queryable relationship metadata concentrated in reusable archetypes and templates.

CONCEPTUAL MODEL

Each clinical entry is anchored in a care contact, assigned to an Episode of Care and classified by a clinical section.

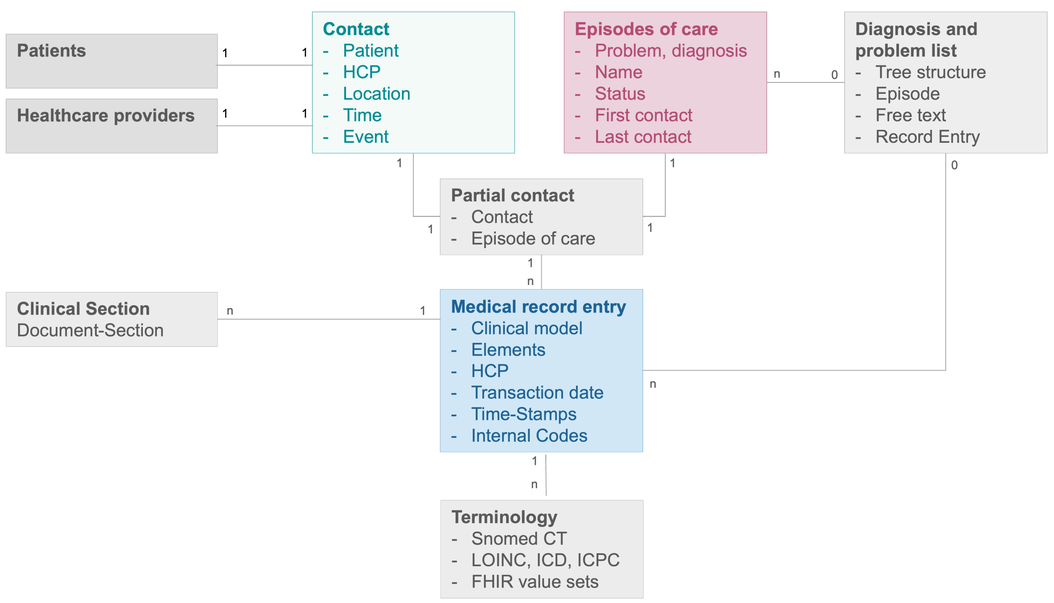


Figure 1. Conceptual data model of the episode-oriented medical record.

openEHR MAPPING

Existing openEHR structures preserve both event context and longitudinal problem context without modifying the Reference Model.

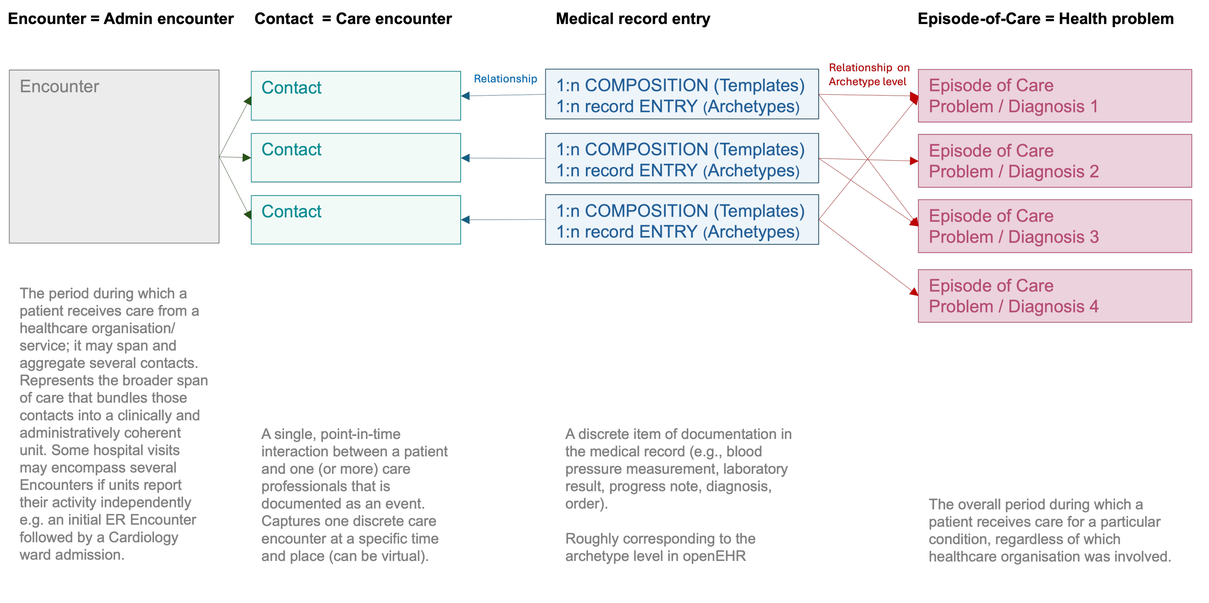


Figure 2. openEHR mapping of contacts, medical record entries and episodes of care.

RECOMMENDED HYBRID

- Episode COMPOSITIONs – longitudinal source of truth
- CLUSTERs – authoritative semantic representation
- FOLDERs – patient-wide navigation
- LINKs – selective explicit relationships

CLINICAL RELEVANCE

- Follow the course of a health problem over time
- Navigate between chronological, episode-centred and problem-list views
- Maintain consistent information without duplicating data

KEY CONTRIBUTION

Episode, diagnosis, problem and past medical history lists are derived clinical views over the same underlying clinical information rather than independently maintained data structures.

CONCLUSION

- Solon's episode-oriented medical record can be represented in openEHR without changing the Reference Model.
- A pragmatic hybrid of CLUSTERs, FOLDERs and LINKs supports longitudinal, problem- and episode-centred records while preserving consistency, provenance, queryability and vendor-neutral persistence.

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Thesis & models

Episode concept

openEHR implementation